

Electromagnetic Waves for Six Marks

Chapter 8 · Electromagnetic Waves — 6 marks. Source: published page 762322ac-2430-4ee0-99aa-22a0cb8e6921. Plain-text maths, not KaTeX. Physics, Derived has one Chapter 8 derivation (PD38, displacement current); the other four here appear nowhere else. This page also adds the theory, the spectrum table, the formula strip and the question tiers.

Scope. No scope cut. Five derivations only — the rest of the chapter is recall, and the spectrum table is the single most reliably asked thing in it.

01 · Theory

1 · Displacement current

Definition to write: displacement current is the current that exists in a region wherever the electric flux through that region is changing with time, given by $I_d = \epsilon_0 (d\Phi_E/dt)$. It is not a flow of charge, and it needs no conductor — it exists in vacuum.

Two situations the examiner asks you to separate:

- **Conduction current, no displacement current** — a steady current in a wire. The electric field in the wire is constant, so the flux does not change.
- **Displacement current, no conduction current** — the space between the plates of a capacitor while it charges or discharges. No charge crosses the gap, but the field between the plates is growing, so the flux is changing.

Its whole point is continuity: at every instant I_d between the plates equals I_c in the wire, so the total current is unbroken all the way round the circuit and Kirchhoff's junction rule survives. This is also why a galvanometer in series with a capacitor kicks momentarily while the capacitor charges on dc, and why current flows continuously when the source is ac.

2 · Characteristics of electromagnetic waves

- They are **transverse**: E and B are perpendicular to each other and both perpendicular to the direction of propagation, which lies along $E \times B$.
- E and B are **in phase** — they reach their maxima and their zeros together — and their magnitudes are locked by $E_0/B_0 = c$.
- They need **no material medium** and travel through vacuum at $c = 3 \times 10^8 \text{ m s}^{-1}$, the same speed for every wavelength.
- They are **electrically neutral**, so electric and magnetic fields do not deflect them.
- They show reflection, refraction, interference, diffraction and **polarisation** — the last one is the direct evidence that they are transverse.
- They carry **energy and momentum**, so they exert pressure on any surface that absorbs them.
- In a medium the speed falls to $v = 1/\sqrt{\mu\epsilon} = c/n$; the frequency stays as the source set it, so the wavelength shortens.

Sunlight warming your hand is the everyday evidence for energy; the photoelectric effect is the laboratory evidence. You do not feel the pressure because $p = U/c$ and c is enormous.

3 • The electromagnetic spectrum

Band	Wavelength	Frequency	How it is produced	One use
γ -rays	$< 10^{-10}$ m	$> 3 \times 10^{18}$ Hz	Radioactive decay of nuclei; nuclear reactions	Destroying cancer cells
X-rays	$10^{-10} - 10^{-8}$ m	$3 \times 10^{16} - 3 \times 10^{18}$ Hz	Fast electrons stopped by a heavy metal target such as tungsten	Detecting bone fractures
Ultraviolet	$10^{-8} - 4 \times 10^{-7}$ m	$7.5 \times 10^{14} - 3 \times 10^{16}$ Hz	Very hot bodies; the Sun; mercury-vapour arcs and special lamps	Sterilising water in purifiers; LASIK eye surgery
Visible light	$4 \times 10^{-7} - 7.5 \times 10^{-7}$ m	$4 \times 10^{14} - 7.5 \times 10^{14}$ Hz	Electrons rearranging in atoms; incandescent bodies	Vision, photography
Infrared	$7.5 \times 10^{-7} - 10^{-3}$ m	$3 \times 10^{11} - 4 \times 10^{14}$ Hz	Vibration of hot bodies, atoms and molecules	Physiotherapy; TV remote controls
Microwaves	$10^{-3} - 10^{-1}$ m	$3 \times 10^9 - 3 \times 10^{11}$ Hz	Klystrons, magnetrons and Gunn diodes	Radar for aircraft navigation; microwave ovens
Radio waves	$> 10^{-1}$ m	$< 3 \times 10^9$ Hz	Accelerated charges in aerials and oscillating LC circuits	Radio and television broadcasting

Order to be able to recite in both directions: γ , X, UV, visible, IR, microwave, radio — increasing wavelength, decreasing frequency. Visible light alone runs 400 nm (violet) to 750 nm (red).

4 • How electromagnetic waves are produced

A charge at rest makes only a static electric field. A charge in uniform motion makes a steady magnetic field as well, but neither field changes, so nothing radiates. Only an **accelerated** charge radiates.

The mechanism to write out: an oscillating charge produces an oscillating electric field in the space around it; that changing electric field produces an oscillating magnetic field; that changing magnetic field produces a further oscillating electric field, and so on. Each field regenerates the other, so the disturbance carries itself outwards through empty space as an electromagnetic wave.

The frequency of the wave equals the frequency of oscillation of the charge. An LC circuit oscillating at $1/(2\pi\sqrt{LC})$ is therefore a source of electromagnetic waves at that frequency, and the energy radiated comes out of the source.

A charge moving in a circle is accelerating — its direction is changing — so it radiates too.

5 • The atmosphere and the spectrum

- **Earth's warmth.** The Sun's visible and short-wave radiation reaches the ground and warms it. The ground re-radiates at longer infrared wavelengths, and greenhouse gases such as CO₂ and water vapour absorb and trap that infrared. Without an atmosphere there would be no such trapping, so the average surface temperature would be *lower* than it is now.
- **The ozone layer.** Ozone on top of the stratosphere absorbs the Sun's harmful ultraviolet before it reaches the surface. UV damages skin and eyes and causes skin cancer, so this thin layer is what makes life on land possible.
- **X-ray astronomy needs satellites.** The atmosphere absorbs X-rays completely, so an X-ray telescope on the ground would see nothing. Visible light and radio waves pass through, which is why optical and radio telescopes work from the ground.
- **Long-distance radio uses short-wave bands.** Short waves are reflected back to Earth by the ionosphere, so a signal can be bounced over the horizon to a distant receiver instead of escaping into space.
- **Radar uses microwaves.** Their short wavelength gives good directionality and resolution, and they pass through the atmosphere with little diffraction and little absorption.

02 • Derivations

D1 • Displacement current — why Ampere's law was incomplete

A parallel-plate capacitor of plate area A is being charged, so a time-varying current $I(t)$ flows in the connecting wire. Draw a circular loop around the wire, and consider two *different* surfaces that share that same loop as their boundary.

1. Ampere's circuital law: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$, where I is the current through any surface bounded by the loop
2. Surface A: a flat disc, pierced by the connecting wire
3. Current through surface A = $I(t)$
4. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I(t) \dots (i)$
5. Surface B: a pot-shaped surface with the same boundary loop, dipping between the capacitor plates and touching no wire
6. No charge crosses the gap between the plates, so the current through surface B = 0
7. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \times 0 = 0 \dots (ii)$
8. The left-hand side of (i) and (ii) is the same integral over the same loop
9. (i) and (ii) contradict each other, so Ampere's law in this form is incomplete
10. Between the plates the field is $E = \sigma/\epsilon_0$
11. Surface charge density $\sigma = Q/A$, Q being the charge on a plate
12. $E = Q/(\epsilon_0 A)$
13. Electric flux through surface B: $\Phi_E = E A$
14. $\Phi_E = [Q/(\epsilon_0 A)] \times A$
15. $\Phi_E = Q/\epsilon_0$
16. $Q = \epsilon_0 \Phi_E$

17. Differentiate with respect to time: $dQ/dt = \epsilon_0 (d\Phi_E/dt)$
18. dQ/dt is the rate at which charge piles up on the plate, which equals the conduction current I_c in the wire
19. $I_c = \epsilon_0 (d\Phi_E/dt)$
20. So a quantity $\epsilon_0 (d\Phi_E/dt)$ exists in the gap with exactly the value of the missing current
21. Maxwell named it the displacement current: $I_d = \epsilon_0 (d\Phi_E/dt)$

Result: $I_d = \epsilon_0 (d\Phi_E/dt)$, and $I_d = I_c$ at every instant

Diagram: capacitor plates with a wire entering from the left. Draw the flat disc surface A cutting the wire, and the pot-shaped surface B bulging round the left plate so its mouth is the same circular loop but its bottom lies in the gap. Mark $I(t)$ on the wire and the electric field lines between the plates.

D2 • The Ampere-Maxwell law

Ampere's law repaired by adding the displacement current found in D1. Quote D1's boxed result first — this derivation is only four lines of algebra on top of it.

Shared setup: D1 and D2 are one continuous argument, and CBSE often asks them as a single question. Learn them as one block: D1 finds the missing term, D2 puts it into the law.

1. From D1: surface A carries conduction current I_c and no changing flux
2. From D1: surface B carries no conduction current but a changing electric flux
3. For the law to give the same answer for both, the total current must be $I = I_c + I_d$
4. $I_d = \epsilon_0 (d\Phi_E/dt)$
5. $I = I_c + \epsilon_0 (d\Phi_E/dt)$
6. Put this total current into $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$
7. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 [I_c + \epsilon_0 (d\Phi_E/dt)]$
8. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_c + \mu_0 \epsilon_0 (d\Phi_E/dt)$
9. Check on surface A: $d\Phi_E/dt = 0$, so $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_c$
10. Check on surface B: $I_c = 0$, so $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \epsilon_0 (d\Phi_E/dt) = \mu_0 I_c$ — (using step 19 of D1)
11. Both surfaces now give the same value of $\oint \mathbf{B} \cdot d\mathbf{l}$ — the contradiction is gone

Result: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 [I_c + \epsilon_0 (d\Phi_E/dt)]$

What to say after the box: this is one of Maxwell's four equations. The new term states that a changing electric field produces a magnetic field — the mirror image of Faraday's law, where a changing magnetic field produces an electric field. That symmetry is what makes a self-sustaining electromagnetic wave possible.

D3 • Speed of electromagnetic waves, and $E_0/B_0 = c$

Free space: no free charges and no conduction current, so $I_c = 0$ everywhere. Only two of Maxwell's equations then matter, and they feed each other.

1. Faraday's law: $\oint \mathbf{E} \cdot d\mathbf{l} = -(d\Phi_B/dt)$ — a changing B makes an E
2. Ampere-Maxwell law from D2 with $I_c = 0$: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \epsilon_0 (d\Phi_E/dt)$ — a changing E makes a B

3. Each field regenerates the other, so the disturbance propagates with no medium
4. Maxwell solved these two together for a plane wave and obtained a wave speed $c = 1/\sqrt{\mu_0\epsilon_0}$
5. $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
6. $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
7. $\mu_0 \epsilon_0 = (4 \times 3.14 \times 10^{-7})(8.85 \times 10^{-12})$
8. $\mu_0 \epsilon_0 = 1.112 \times 10^{-17}$
9. $\sqrt{\mu_0 \epsilon_0} = 3.335 \times 10^{-9}$
10. $c = 1 / (3.335 \times 10^{-9})$
11. $c = 3.00 \times 10^8 \text{ m s}^{-1}$
12. This equals the measured speed of light, which is how light was identified as an electromagnetic wave
13. For a plane wave along z: $E = E_0 \sin(kz - \omega t)$ and $B = B_0 \sin(kz - \omega t)$
14. Both carry the same k and the same ω , so both travel at the same speed ω/k
15. $\omega/k = c$
16. Maxwell's equations further require the amplitudes to satisfy $E_0 = c B_0$
17. $E_0 / B_0 = c$
18. In a medium of permeability μ and permittivity ϵ : $v = 1/\sqrt{\mu\epsilon}$
19. $v = 1/\sqrt{(\mu_r \mu_0 \epsilon_r \epsilon_0)} = c/\sqrt{(\mu_r \epsilon_r)} = c/n$

Result: $c = 1/\sqrt{\mu_0\epsilon_0} = 3 \times 10^8 \text{ m s}^{-1} \cdot E_0/B_0 = c \cdot v = c/\sqrt{(\mu_r \epsilon_r)}$

Step 4 is the one place a link is quoted rather than worked. That is deliberate: solving the coupled equations needs the wave equation, which is outside Class XII. What CBSE actually asks is "show that $1/\sqrt{(\mu_0\epsilon_0)}$ gives the velocity of an EM wave in free space" — that is steps 5 to 12, the numerical substitution, and you should write every one of them out.

D4 • Transverse nature, and the orientation of E, B and propagation

A plane electromagnetic wave travelling along the z-axis in free space, written as $E_x = E_0 \sin(kz - \omega t)$ and $B_y = B_0 \sin(kz - \omega t)$.

1. Gauss's law for electricity in free space: $\oint E \cdot dA = q/\epsilon_0 = 0$ — (no charge enclosed)
2. Take a small box with two faces perpendicular to the propagation direction z
3. A component of E along z would send flux in through one face and out through the other, unequally as the wave passes
4. That would give a non-zero net flux, contradicting step 1
5. So E has no component along z
6. Gauss's law for magnetism: $\oint B \cdot dA = 0$ always — (no magnetic monopoles)
7. By the identical argument, B has no component along z
8. Both E and B lie entirely in the plane perpendicular to the propagation direction — the wave is transverse
9. Faraday's law links the B along y to an E that varies along z and points along x
10. The Ampere–Maxwell law links the E along x to a B that varies along z and points along y

11. So E is along x , B is along y , propagation is along z : $E \perp B$, and both $\perp$ propagation
12. $\hat{x} \times \hat{y} = \hat{z}$, so the direction of propagation is along $E \times B$
13. E , B and the propagation vector form a right-handed set
14. Both fields carry the same phase ($kz - \omega t$), so they reach their maxima and their zeros at the same instant — E and B are in phase

Result: $E \perp B \perp$ propagation; propagation along $E \times B$; phase difference between E and B is zero

Diagram: three axes with z pointing right. Draw the E sinusoid oscillating up and down in the x - z plane and the B sinusoid oscillating in and out in the y - z plane, both crossing zero at the same points along z . Label the propagation arrow along $+z$.

Polarisation is the experimental proof of this: only a transverse wave can be polarised, and light can be.

D5 • Energy density and intensity

The same plane wave, $E = E_0 \sin(kz - \omega t)$ and $B = B_0 \sin(kz - \omega t)$, travelling through free space.

1. Energy density of an electric field: $u_E = \frac{1}{2} \epsilon_0 E^2$
2. Energy density of a magnetic field: $u_B = B^2 / (2\mu_0)$
3. From D3: $E = cB$
4. Substitute in step 1: $u_E = \frac{1}{2} \epsilon_0 (cB)^2$
5. $u_E = \frac{1}{2} \epsilon_0 c^2 B^2$
6. From D3: $c = 1/\sqrt{(\mu_0\epsilon_0)}$, so $c^2 = 1/(\mu_0\epsilon_0)$
7. $u_E = \frac{1}{2} \epsilon_0 B^2 \times 1/(\mu_0 \epsilon_0)$
8. $u_E = B^2 / (2\mu_0)$
9. Comparing with step 2: $u_E = u_B$
10. Total energy density $u = u_E + u_B = \frac{1}{2} \epsilon_0 E^2 + B^2/(2\mu_0)$
11. Since the two halves are equal: $u = 2 \times (\frac{1}{2} \epsilon_0 E^2) = \epsilon_0 E^2$
12. Average over one cycle: $\langle \sin^2(kz - \omega t) \rangle = \frac{1}{2}$
13. $\langle u_E \rangle = \frac{1}{2} \epsilon_0 E_0^2 \times \frac{1}{2} = \frac{1}{4} \epsilon_0 E_0^2$
14. $\langle u_B \rangle = [B_0^2/(2\mu_0)] \times \frac{1}{2} = B_0^2/(4\mu_0) = \frac{1}{4} \epsilon_0 E_0^2$ — (by step 8)
15. $\langle u \rangle = \langle u_E \rangle + \langle u_B \rangle = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4} \epsilon_0 E_0^2$
16. $\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2$
17. $E_{rms} = E_0/\sqrt{2}$, so $E_0^2 = 2 E_{rms}^2$
18. $\langle u \rangle = \frac{1}{2} \epsilon_0 (2 E_{rms}^2) = \epsilon_0 E_{rms}^2$
19. Intensity I = energy crossing unit area per unit time
20. In time t the wave sweeps a cylinder of cross-section A and length ct
21. Energy in that cylinder = $\langle u \rangle \times (A c t)$
22. $I = \text{energy} / (A t) = \langle u \rangle A c t / (A t)$
23. $I = \langle u \rangle c$
24. $I = \frac{1}{2} \epsilon_0 E_0^2 c$

Result: $u_E = u_B \cdot \langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = \epsilon_0 E_{\text{rms}}^2 \cdot I = \frac{1}{2} \epsilon_0 E_0^2 c$

Momentum follows: a wave delivering energy U to a fully absorbing surface delivers momentum $p = U/c$, so it presses on that surface. The pressure is tiny only because c is so large.

03 · Formula strip

● *must be instant* · ○ *a few seconds is fine*

● Displacement current

$I_d = \epsilon_0 (d\Phi_E/dt)$ — Φ_E = electric flux in V m (or $\text{N m}^2 \text{C}^{-1}$), $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$.
 I_d in ampere (A). Equal to the conduction current at every instant.

● Ampere-Maxwell law, the full corrected form

$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 [I_c + \epsilon_0 (d\Phi_E/dt)]$ — B in tesla (T), dl in metre (m), I_c = conduction current in ampere (A), $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$.

● Speed of electromagnetic waves in free space

$c = 1/\sqrt{\mu_0 \epsilon_0} = 3 \times 10^8 \text{ m s}^{-1}$ — same for every wavelength. This equality with the measured speed of light is what identified light as an EM wave.

● Ratio of the electric and magnetic field amplitudes

$E_0 / B_0 = c$ — E_0 in V m^{-1} (or N C^{-1}), B_0 in tesla (T). Also holds instant by instant: $E = cB$.

● Speed in a material medium

$v = 1/\sqrt{\mu\epsilon} = c/\sqrt{\mu_r \epsilon_r} = c/n$ — v in m s^{-1} ; μ_r , ϵ_r and n are dimensionless. The frequency is unchanged, so the wavelength shortens by the factor n .

● Frequency, wavelength and speed

$c = v \lambda$ — ν in hertz (Hz), λ in metre (m). The workhorse of every spectrum numerical.

○ Wave number and angular frequency of a written wave

$E = E_0 \sin(kz - \omega t)$, $k = 2\pi/\lambda$, $\omega = 2\pi\nu$, $c = \omega/k$ — k in rad m^{-1} , ω in rad s^{-1} . Read λ and ν straight off the coefficients of z and t .

● Energy densities of the electric and magnetic parts

$u_E = \frac{1}{2} \epsilon_0 E^2$ · $u_B = B^2/(2\mu_0)$ — both in joule per cubic metre (J m^{-3}). In an EM wave these two are equal at every point.

○ Average total energy density of an EM wave

$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = \epsilon_0 E_{\text{rms}}^2$ — in J m^{-3} . E_0 = amplitude, $E_{\text{rms}} = E_0/\sqrt{2}$, both in V m^{-1} .

○ Intensity of an electromagnetic wave

$I = \langle u \rangle c = \frac{1}{2} \epsilon_0 E_0^2 c$ — in watt per square metre (W m^{-2}). Energy crossing unit area per unit time.

○ Momentum delivered by a wave that is fully absorbed

$p = U / c$ — U = energy absorbed in joule (J), p in kg m s^{-1} . Halve nothing — for a fully reflecting surface the momentum transferred is $2U/c$.

○ Energy of one photon of the radiation

$E = h\nu = hc/\lambda$ — $h = 6.63 \times 10^{-34} \text{ J s}$. E in joule; divide by 1.6×10^{-19} to get electronvolt (eV).

● The two constants, with units

$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$ · $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ — permeability and permittivity of free space. You will be asked to substitute these into $c = 1/\sqrt{\mu_0\epsilon_0}$.

● Wavelength range of visible light, in nanometre

400 nm (violet) to 750 nm (red) — $1 \text{ nm} = 10^{-9} \text{ m}$. Anything shorter than 400 nm is ultraviolet, anything longer than 750 nm is infrared.

04 · Questions to attempt

Question numbers only, from Xam Idea Chapter 8 and the NCERT questions reprinted inside it. Section names are Xam Idea's own. At 6 marks this chapter will not carry a five-mark question, so the Tier 1 list is weighted towards the 3-mark slots.

Tier 1 — must do

15 questions · the 3-mark slots and the spectrum recall

Question	Page	Why
Short Answer Q10	p. 279	Ampere generalised, full
Short Answer Q9	p. 278	Same argument, shorter
Short Answer Q3	p. 276	Conduction versus displacement
Short Answer Q4	p. 276	Current stays continuous
Short Answer Q5	p. 277	Diagram plus the c substitution
Short Answer Q1	p. 275	Spectrum table one
Short Answer Q2	p. 275	Spectrum table two
Short Answer Q6	p. 277	Identify plus production
Short Answer Q11	p. 280	Energy, momentum, density
Short Answer Q14	p. 281	Radar, atmosphere, pressure
Short Answer Q16	p. 281	Cancer, ozone, momentum
Short Answer Q17	p. 282	Four bands, ordered
Very Short Ans Q7	p. 273	The corrected law, stated
NCERT Q10	p. 266	Proves u_E equals u_B
Practice Q11	p. 284	Same proof, asked again

Tier 2 — if time

extra pattern coverage, mostly 1 and 2 marks

Question	Page	Why
NCERT Q1, Q2	p. 264-265	Displacement current numericals
NCERT Q4, Q7, Q8	p. 265-266	E_0 , B_0 , ω , k drills
NCERT Q3, Q5, Q6	p. 265	$c = \nu\lambda$ one-liners
MCQ 1, 2, 3	p. 267	Phase, free space, ratio
MCQ 23-27	p. 270	Density, uses, displacement
MCQ 4-22	p. 268-269	Rest of the set, fast
Assertion-Reason 1-10	p. 270-271	Whole set, ten minutes
Case Study (i)-(iv)	p. 272	Order, transverse, wave number
Very Short Ans Q1-Q6	p. 272-273	Two-mark definitions
Very Short Ans Q8-Q14	p. 274-275	Spectrum recall, rapid fire
Short Answer Q7, Q8	p. 277-278	Ordering, dc versus ac
Short Answer Q12, Q13	p. 280	Oven, radar, sources
Practice Q1 (i)-(x), Q2	p. 283	One-markers plus assertion
Practice Q3, Q4, Q5, Q9	p. 283	Short numericals

Tier 3 — skip unless revising

off the blueprint, or already done above

Belongs to a different chapter: NCERT Q9 (p. 266) builds a table of photon energies across the spectrum from $E = h\nu$. That is Dual Nature material, not this unit's six marks.

Duplicates — do one of each pair, not both: NCERT Q10 $\equiv$ Practice Q10. Short Answer Q15 $\equiv$ Short Answer Q16, same three-part shape. Practice Q12 $\equiv$ Short Answer Q10. Practice Q14 $\equiv$ Short Answer Q15. Practice Q6, Q7, Q8, Q13 are all restatements of Very Short Answer and Short Answer questions already in Tiers 1 and 2.

Read the answer, do not write it out: at 6 marks the chapter cannot take more than about four questions on the paper. Once Tier 1 is done, the return on writing out any further recall answer is close to zero — spend the time on Chapter 7 instead.

The five numerical types this chapter can ask

1. **Displacement current between the plates of a charging capacitor** — $I_d = \epsilon_0 (d\Phi_E/dt) = \epsilon_0 A (dE/dt)$, and $I_d = I_c$
2. **Move between frequency and wavelength anywhere on the spectrum** — $c = \nu\lambda$
3. **One field amplitude from the other** — $E_0/B_0 = c$
4. **Read λ , ν , ω and speed off a wave written as an equation** — $k = 2\pi/\lambda$, $\omega = 2\pi\nu$, $c = \omega/k$
5. **Energy density, intensity or momentum of a beam** — $\langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2$, $I = \langle u \rangle c$, $p = U/c$